Experiment : 6

Date:

USE CASE DIAGRAM

Aim:

To design and model the functional requirements of an Automated Teller Machine (ATM) Withdrawal System using a UML Use Case Diagram and to understand the interaction between actors and system functionalities using Rational Rose.

Introduction:

Unified Modeling Language (UML) is a standardized modeling language used for specifying, visualizing, constructing, and documenting the components of a software system. Among various UML diagrams, the Use Case Diagram is widely used during the requirement analysis phase.

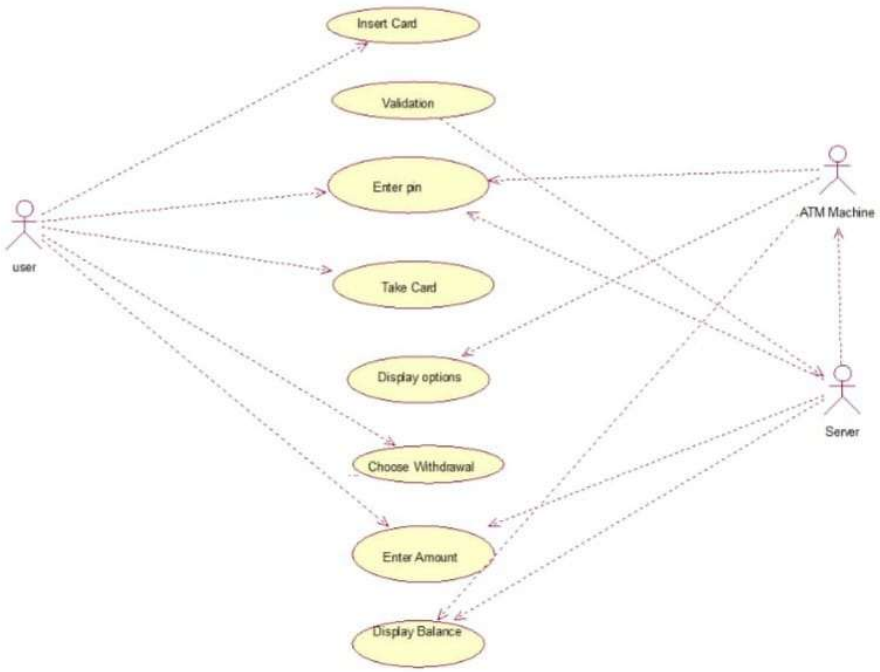
A Use Case Diagram provides a high-level view of the system by representing how external entities, called actors, interact with the system to achieve specific goals. It focuses mainly on what the system does rather than how the system works internally.

In this experiment, an ATM Withdrawal System is modeled using a Use Case Diagram to clearly represent user interactions, transaction flow, and system responsibilities.

Use Case Diagram:

A Use Case Diagram is a UML diagram that describes the functional behavior of a system by showing actors, use cases, and their relationships. It helps in

identifying system requirements from the user's perspective and simplifies system understanding by providing a graphical representation of interactions.



Software/Tools Required:

- Rational Rose UML Modeling Tool
- Windows Operating System

Actors Identified:

1. **User**
   The person who operates the ATM machine to perform transactions such as inserting the card, entering PIN, and withdrawing cash.

2. **ATM Machine**
   The interface that allows the user to interact with the banking system and perform operations.
3. **Server**
   The banking system that validates user details, processes transactions, and updates account information.

Use Cases Identified:

- Insert Card
- Validation
- Enter PIN
- Display Options
- Choose Withdrawal
- Enter Amount
- Display Balance
- Take Card

Each use case represents a specific functionality provided by the ATM system.

Description of the System:

The ATM Withdrawal System begins when the user inserts the ATM card into the machine. The ATM Machine communicates with the server to validate the card information. After validation, the system prompts the user to enter a PIN number.

Once the PIN is entered, the server verifies the authentication details. If the verification is successful, the ATM displays a list of available transaction options such as withdrawal, balance inquiry, and others. The user selects the withdrawal option and enters the required amount.

The ATM Machine sends the transaction request to the server, which checks the account balance and approves the transaction. After approval, the ATM dispenses cash to the user and displays the remaining balance on the screen. Finally, the ATM returns the card to the user, completing the transaction process.

Relationships Observed:

- **Association Relationship:** Connects actors with use cases to represent interaction.
- **Communication:** ATM Machine communicates with the Server for validation and processing.
- **Sequential Flow:** The process follows a step-by-step order from card insertion to transaction completion.

Observation:

The ATM Withdrawal System was successfully analyzed using a UML Use Case Diagram. The diagram clearly identifies three main actors: User, ATM Machine, and Server. Various use cases such as Insert Card, Validation, Enter PIN, Display Options, Choose Withdrawal, Enter Amount, Display Balance, and Take Card were included to represent system functionality hookup.

The interactions between actors and use cases were illustrated through associations, which helped in understanding the flow of activities within the system. The ATM Machine acts as an intermediate interface, while the Server performs authentication and transaction processing. The Use Case Diagram provides a clear overview of how the user interacts with the system and how each function is executed sequentially.

Advantages of Use Case Diagram:

- Easy to understand and simple to represent.
- Helps in requirement analysis and system planning.
- Clearly identifies actors and system functionalities.
- Improves communication between developers and users.

Conclusion:

Thus, the ATM Withdrawal System was successfully designed and represented using a UML Use Case Diagram in Rational Rose. The diagram effectively shows actors, use cases, system interactions, and functional workflow of the ATM system.